SHUVAM SINGH

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EDUCATION

SRM University - AP Andhra Pradesh, India

*Bachelor of Technology in Computer Science Engineering **(Big Data Specialization)** — Current CGPA: 8.61* *2022 – 2026*

- Relevant Coursework: Big Data Analytics, ML, Distributed Systems, Cloud Computing, Software Engineering, UI.
- Research Interest: Recommendation Systems, Commercial Aviation Systems, Adaptive AI, RAG, Data Science, SNA.

EXPERIENCE

Systems Intern & Technical Associate May–June 2024 | Dec 2025 – Present

United Lubricants & SatyaDip International *Lalitpur, Nepal*

- Contributed to digital transformation and ERP migration (legacy CRM to Odoo), alongside managing IT infrastructure, networking and workflow automation across manufacturing and multi branch operations.
- Engineered a centralized **Smart Inventory System** and automated QR based stock tracking workflows, bridging physical manufacturing stages with digital records to optimize replenishment and reduce costs.
- Leveraged **LLMs and data analytics** to process 100+ HS import codes and retail sales data, developing product intelligence frameworks that directly guided EV manufacturing expansion and sourcing strategies.
- Designed and deployed a locally hosted **RAG based internal knowledge system** and developed corporate web platforms featuring real time, region based stock availability and AI generated marketing assets.

PROJECTS

Dynamic Template Guided Segmented Generation | *Python, RAG, Ollama, Llama.cpp, ChromaDB* **Jun. 2026**

- Designed a deterministic **context orchestration architecture** for long form generation on resource constrained LLMs using dynamic template planning and segment-aware retrieval.
- Developed **Iterative Full Window Injection** and structured graph state memory to maximize context utilization while maintaining accurate multi step synthesis under strict token budgets.
- Engineered an end to end pipeline integrating adaptive retrieval, evidence validation, token budget management and citation aware response generation, achieving a **203% improvement** over a naive RAG baseline in a custom 500 query benchmark

Self Evolving Cognitive Architecture & Bio Memory Model | *LLMs, LoRA, ChromaDB, PEFT* **Jan. 2026**

- **Accepted for Presentation** at the **IEEE 4th International Conference on AI Innovation (ICAII 2026)**
- Designed a modular lifelong learning architecture integrating **semantic vector memory**, **knowledge graphs** and **LoRA based parameter efficient continual learning** for adaptive personalization without full model retraining.
- Implemented biologically inspired memory management using the **Ebbinghaus forgetting curve**, retrieval induced reinforcement, and sleep phase consolidation for intelligent memory retention and pruning.
- Built an **asynchronous hallucination validation pipeline** with real time web verification, confidence based feedback loops & optimized inference on consumer grade hardware.

AI Hallucination Mitigation | *Python, Ranking Algorithms, NLP, LLM, Multimodal* **Dec. 2025**

- Conducted undergraduate research focused on reducing **AI hallucination** through enhanced query processing, page ranking algorithms and Multimodal concepts.
- Implemented multiple validation layers to filter inconsistent or conflicting outputs and extract relevant information around user queries.

Advanced RAG System with Multi Model Validation | *OpenAI Whisper, Qwen Vision, Mistral, LanceDB, Python* **Sep. 2025**

- Engineered a **Smart RAG** system utilizing the entire web as a database with advanced filtering mechanisms.
- Implemented a **Policy based access control** using vector storage to reduce context drift.

Smart Hospital ERP with LLM Integration | *Python, LLM APIs, SQL* **Apr. 2025**

- Built a Hospital ERP system enhanced with **LLM** for automated patient query handling and record summarization.
- Streamlined administrative workflows and improved data accessibility for medical staff and patients.

Gesture & Emotion Based Smart Attendance | *OpenCV, Deep Learning, Python* **Feb. 2025**

- Developed an attendance system that responds to **hand gestures & facial emotions** based on multiple validation layers.
- Gesture controlled system automated class tasks with emotion based implicit productivity feedback.

Air Traffic Management & Aerospace Intelligence | *Python, XGBoost, GeoPandas, Sklearn, Streamlit* **Oct. 2024**

- **Accepted for Presentation** at the **International Conference on Advances in Aerospace Technologies (ICAAsT 2024)**.
- Designed a 4D trajectory conflict detection and route optimization system using **GeoPandas** and spatial temporal waypoint modeling under weather, airspace congestion and restricted corridor constraints.
- Engineered an end to end flight safety platform integrating **XGBoost/Random Forest** risk classifiers with predictive maintenance ML to generate real time dispatch directives on an interactive **Streamlit** dashboard.

Advanced Research & Future Initiatives

AI Driven eVTOL & UAM Air Traffic Control System | *Architecture & Systems Design* **Ongoing**

- Architecting an open, AI native Air Traffic Control framework specifically for drones, quadcopters and eVTOLs, operating independently of legacy commercial aviation systems.
- Designing dynamic, AI optimized waypoint routing algorithms that balance flight efficiency, noise reduction, smart maneuvering and real time traffic and climatic awareness.
- Integrating predictive models for optimal charging station placement and weather adaptive flight path adjustments to ensure safe, scalable Urban Air Mobility (UAM).

Elser: Symbol Based Neural Processing Engine | *Novel Architecture, Compute Optimization* **Exploratory research**

- Designing a **ground up symbolic processing architecture** that represents language through a compact, custom compositional system of **Psifi, Psif, Sif and Keno** units rather than conventional token based representations.
- Developing a bidirectional **natural language** ↔ **Elser** engine for encoding and processing linguistic structure, context, semantics, capitalization, intent and affective information within a dedicated symbolic state space.
- Exploring **deterministic symbolic reasoning and structured state processing** as an alternative or complement to conventional neural computation, with a focus on reducing representation and inference overhead while preserving contextual understanding.

Technical Skills

Languages: Python, SQL, C/C++

AI & LLM Systems: RAG, Context Engineering, LoRA/PEFT, Prompt Engineering, Agentic AI, Knowledge Graphs, Vector Databases, Semantic Search, LLM Inference Optimization, Continual Learning

Frameworks & Tools: Ollama, Llama.cpp, VectorDB

Systems: Workflow Automation, System Integration, AWS Fundamentals

Research Interests: AGI, AI Memory Architectures, Edge AI, Aviation AI, Urban Air Mobility, Symbolic AI, Intelligent Transportation Systems

Professional Skills

Communication: Public Speaking, Technical Presentation, Scientific Writing, Cross cultural Communication (**IELTS Band 7.5**)

Leadership: Team Leadership, Event Management, Project Coordination, Research Collaboration, Strategic Planning

Strengths: Problem Solving, Analytical Thinking, Research & Innovation, Rapid Learning, Technical Documentation

Leadership & Achievements

- 2 research papers accepted, for presentation at the **IEEE Conference on Artificial Intelligence Innovation (ICAII 2026)**, and another on **AI driven Air Traffic Management** accepted at the **International Conference on Advances in Aerospace Technologies (ICAAsT 2024)**.
- Helped to Complete **Phase I** development of an autonomous self driving delivery vehicle that involved world model learning and beta testing on Nepalese landscape.
- Served as a member of the **Student Council**, and active member of Photography club and Volunteered in multiple big events.
- Contributed to business expansion strategy in the **Electric Vehicle Manufacturing** sector through data driven market analysis and strategic planning.